

## CO-03/Assignment-03

Set:-1

Set 1

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(19)

① Consider the following two Hash algorithm :

MD5 and SHA-1. Both are used to produce message digests.

a) Compare the output length of MD5 and SHA-1.

Ans:-

| Feature             | MD5               | SHA-1         |
|---------------------|-------------------|---------------|
| Digest size         | 128 bits          | 160 bits      |
| Hex representation  | 32 characters     | 40 characters |
| Internal block size | 512 bits          | 512 bits      |
| Number of rounds    | 64                | 80            |
| Designed by         | Ron Rivest (1991) | NSA (1995)    |

⇒ A Hash function takes an input message of any length and compresses it into a fixed-length output called a message digest. MD5 always output 128 bit's, SHA-1 output a longer 160-bit digest.

MD5 : effort  $\approx 2^{64}$  operationsSHA-1 : effort  $\approx 2^{80}$  operations

∴ SHA-1 Produces a longer hash than MD5.

b) Why MD5 is no longer considered secure.

Ans:- A cryptographic hash function must satisfy 3 properties

1. Pre-image resistance: Hard to find  $M$  given  $H(M)$
2. Second pre-image resistance: Hard to find  $M_2 \neq M_1$  such that  $H(M_1) = H(M_2)$
3. Collision resistance: Hard to find any two message  $M_1 \neq M_2$  such that  $H(M_1) = H(M_2)$

Since MD5 no longer "guarantees" different data  $\rightarrow$  different hash, "it cannot be trusted anywhere an attacker might deliberately try to forge data - digital signatures, SSL/TLS certificates, software integrity checks, Password, Hashing, etc. it is only acceptable today for non-adversarial checksums.

c) Two different files with the same MD5 hash - what attack is this?

Ans: This is called collision attack.

collision attack:- A collision attack occurs when an attacker find two distinct inputs,  $M_1$  and  $M_2$ , such that  $H(M_1) = H(M_2)$

## Types of relevant:

- Classical collision attack:- Attacker has full freedom to choose both  $M_1$  and  $M_2$ , with no control over their content
- Chosen-prefix collision attack:- a stronger, more dangerous version, where the attacker starts with two different, meaning-full prefixes and appends carefully crafted "garbage" bytes so both resulting files hash to the same value.

File-1  $\rightarrow$  MD5 Hash X

File-2  $\rightarrow$  MD5 Hash X

$\therefore$  The attacker exploits this property to make a different file appear to have same hash as the original file.

d) Why is SHA-256 consider more secure than SHA-1? Illustrate with reference resistance:

Ans:-

(i) Large output space:

| Algorithm | Digest size | Brute-force collision cost  |
|-----------|-------------|-----------------------------|
| SHA-1     | 160 bits    | $\approx 2^{80}$ operation  |
| SHA-256   | 256 bits    | $\approx 2^{128}$ operation |

SHA-1 also has real cryptanalytic weakness - proven in 2017 by Google/CWI's "SHAttered" attack (two pdf's some SHA-1 hashes).  
SHA-256 has a larger output and stronger internal design with no known practical collision attack, making it the current standard for TLS/certificates/signature.

∴ SHA-256 is preferred over SHA-1 for modern security application.